CS24307 — THEORY (3.0 CR) MODULE II NOTES BIT MESRA | NEP SCHEME

Module 2: Search Algorithms & Game AI

BFS, DFS, UCS, A* Admissibility, Minimax & Alpha-Beta Pruning

Module II: Problem Solving by Search & Game AI — Topics Covered

1. Uninformed Search: BFS, DFS, UCS, IDS
2. Heuristic Informed Search: Greedy & A*
3. Admissibility and Consistency of Heuristics
4. Local Search: Hill Climbing & Simulated Annealing
5. Adversarial Search: Minimax & Alpha-Beta Pruning

1. A^* Search Algorithm & Heuristic Optimality

A^* evaluates nodes by: $f(n) = g(n) + h(n)$, where $g(n)$ is the true path cost from start to n, and $h(n)$ is estimated cost from n to goal.

Theorems of A^* Optimality

Admissibility ($h(n) \leq h^*(n)$): A heuristic is admissible if it never overestimates the true cost to reach the goal. Admissibility guarantees that A^* tree search is optimal.
Consistency (Monotonicity): $h(n) \leq c(n, a, n') + h(n')$. Consistency guarantees that A^* graph search is optimal without reopening closed nodes.

2. Minimax with Alpha-Beta Pruning

α is the highest value found so far for MAX; β is the lowest value found so far for MIN. **Pruning Rule:** Whenever $\alpha \geq \beta$, prune the remaining subtrees below the current node.